



CCNA 200-301 v2.0

Weekly Lecture Schedule • Fall 2026

Cisco Certified Network Associate • Notes, Configurations and Labs • written from zero, no prior networking assumed

FORMAT 16 weeks, two sessions a week: theory and lab	EXAM 200-301, 120 minutes; v2.0 live from 3 February 2027	HANDS-ON 55% of the blueprint is configure, troubleshoot or diagnose
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📅	📖 Theory session: chapters	🔧 Lab session	📋 Assessment
1	1 Networks, Models and This Exam – layers, devices, the exam 2 The Physical Layer – media, cable faults, interface counters	Cabling, show interfaces, injected cable faults	—
2	3 Ethernet and Switching – hex, the frame, how a switch learns 7 The IOS Command Line – modes, running vs startup, first setup	First switch: hostname, SSH, save	—
3	4 IPv4 Addressing and Subnetting – binary, the mask, the block-size method, subnetting to a requirement	Subnetting drills (Appendix C)	Quiz 1 — subnetting, timed
4	5 IPv6 Addressing – address types, EUI-64, configuring IPv6 6 TCP, UDP and the Diagnostic Toolkit – ports, ping, traceroute	Dual-stack a small network; extended ping and traceroute	—
5	8 VLANs and Access Ports – what a VLAN does, creating, verifying 9 Edge-Host Connectivity – PoE, voice VLANs, APs, IoT devices	VLANs, access ports, voice and PoE	—
6	10 Trunking, 802.1Q and SVIs – tagging, DTP, inter-VLAN routing 11 EtherChannel and LACP – negotiation, layer 2 and layer 3 bundles	Trunk two switches, route between VLANs, bundle links	Lab exam 1 — build a switched network
7	12 Rapid PVST+ and Protection – root election, protection features 13 CDP and LLDP – documenting a network from the devices	Steer the root bridge; document a network from the devices	—
8	14 Troubleshooting Layer 2 – scope, three commands, six faults 33 The Troubleshooting Method – introduced now, used in every lab	Five injected faults against the clock	Mid-term — Parts I and II
9	15 The Routing Table – longest-prefix match, administrative distance 16 Static Routing – four kinds, IPv4 and IPv6, floating statics	Read tables, predict paths, four kinds of static route	—
10	17 OSPFv2, Single Area – adjacencies, configuration, DR and BDR 18 OSPFv3 for IPv6 – what stays the same, what changes	Single-area OSPF, then dual-stack	—
11	19 HSRP and VRRP – the first-hop problem, configuring, status 20 DHCP – client, server and relay roles, troubleshooting	Fail a gateway; DHCP as client, server and relay	Lab exam 2 — routing and first-hop redundancy
12	21 DNS – how a name resolves, record types, diagnosis 22 NAT and PAT – three kinds, four terms, verifying	Diagnose a broken zone; PAT and port forwarding	—
13	23 AAA and Secure Transfer – local access, TACACS+, RADIUS 24 Access Control Lists – wildcard masks, standard, extended, named	AAA with a fallback; write, read and repair ACLs	—
14	25 Layer 2 Security – port security, DHCP snooping, DAI, storm control, RA guard 26 IPsec VPNs – site-to-site vs remote access, tunnel mode	Port security, DHCP snooping, DAI, storm control	Quiz 2 — Part IV
15	27 Wireless Principles – bands, RF, security, interference 28 Client Troubleshooting – host first, three operating systems 29 Hypervisors and Containers – VMs, containers, network impact	A real site survey; the same fault on three operating systems	—
16	30 Management, SNMP and Syslog – five management approaches 31 Ansible – configuration management, inventory, playbooks 32 AI in Network Operations – generative and agentic, prompts 34 Exam Strategy – the last session before the mock	Syslog severities; a playbook across three devices	Final — full mock paper, 120 minutes

SYLLABUS ASSESSMENT WEIGHTS

30% Laboratory portfolio	25% Break & Fix assessments	15% Configuration midterm	20% Theory examination	10% Professional practice
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I Foundations

IV Network Services and Security

II Switching and Network Access

V Wireless, Virtualisation and Operations

III IP Routing

VI Sitting the Exam

Chapter numbers and full titles are in the CCNA v2.0 handout (Appendix G, Schedule, Lab Index and Exam Facts). Labs in Parts I–III run in Packet Tracer; from Part IV they move to CML or GNS3 where real platform behaviour matters. Chapter 33 is used in every lab after Week 8; Chapter 34 belongs in the last session before the mock. Exam domains: 1.0 Network Infrastructure and Connectivity 25%; 2.0 Switching and Network Access 25%; 3.0 IP Routing 20%; 4.0 Network Services and Security 20%; 5.0 AI, Network Operations and Management 10%. Cisco does not publish the question count or the passing score.